MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **DESIGN CALCULATION**
4000 PROCESS PLANT: ABLUTION BUILDING
DESIGN OF BUILDING STRUCTURE

Document Number **MD-216-4000-EG-CV-CAL-1073**
Revision: **D01**
Revision Date: **30.10.2023**
Status: **AS BUILT**

Outotec Document Number:	900707	GAU01	AN03	CED01	40006	OU100687080	2
L&T Document Number:	O19045	900	C	K	24034	001	2

REVISION HISTORY

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
A01	11.05.2020	FIRST ISSUE	MF	AVG	SJM	
C01	15.06.2020	ISSUED FOR CONSTRUCTION	MF	AVG	SJM	
D01	30.10.2023	AS BUILT	MF	AVG	SJM	

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
Typ.	:	Typical
HCS	:	Hollow Core Slab
Fc'	:	Concrete Compressive Strength
Fy	:	Yield Strength of Reinforcing Steel
ϕ	:	Strength Reduction Factor
SBC	:	Saudi Building Code
SASO	:	Saudi Standards, Metrology and Quality Organization
ACI	:	American Concrete Institute
ASCE	:	American Society of Civil Engineers
PCI	:	Precast/Prestressed Concrete Institute

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1. INTENT

This Document is intended to produce the technical needs in Engineering including particular reference to design justification & Engineering validation works. The work described below is related to Design of Ablution Building.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The Scope of this document is limited to Complete Analysis & Structural Design Verification of Ablution Building only.

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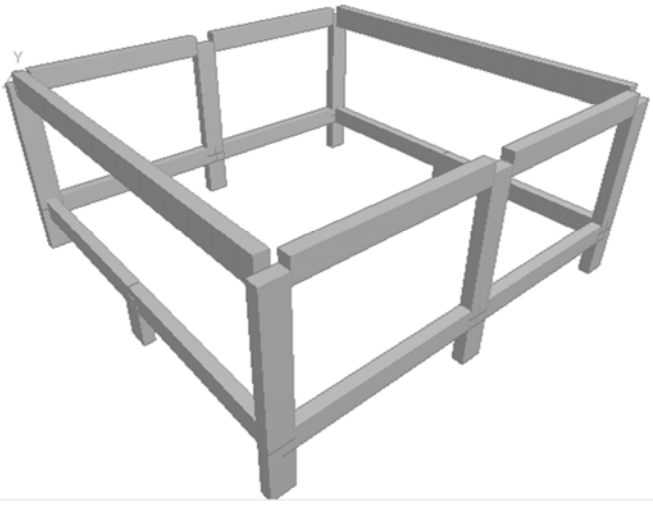
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4. DESIGN METHODOLOGY & ASSUMPTIONS

This Building Is a Single Story Frame Structure with Foundation Being Cast in Site. The Building Shall Have Insulated Block Walls and Cast in Site Framing Beams to support the HCS (Hollow Core Slab)

The Superstructure for this Building was modeled using STAAD software. The Analysis of Frames (Cast-In-Situ Columns & Beams) has been performed meeting design requirements & keeping in view the Architectural requirements. The loads were applied as per SBC/ASCE/ACI standards & analysis was done to generate the Column Reactions at the Base that are assumed to be fixed. The Foundation design has been done by importing Column Base Reactions into Foundation Design Software.



STAAD 3D MODEL

4.1. Codes

[1] SBC-301-2007-Structural Loading & Forces.

[2] SBC-303-2007-Structural Soil & Foundations.

[3] SBC-304-2007-Structural Concrete Structures.

[4] SASO SSA 2 – Steel Bars for the Reinforcement of Concrete.

[5] SASO SSA 224 – Steel Fabric for Reinforcement of Concrete

[6] ASCE-7-2005 Minimum Design Loads for Buildings and other Structures.

[7] ACI 318-08 Building Code Requirements.

[8] PCI –MNL-120-04 –Design Handbook, 6th edition.

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4.2. Geotechnical investigations

Foundations have been designed based on recommendations of the Soil Investigation Report, Doc # MD-216-3000-EG-CV-CAL-1008-A02.

4.3. Material Properties

Compressive Strength of Concrete at 28th days

fc' = 35 MPa (5.00 Ksi) for beams, columns, suspended slabs.

fc' = 35 MPa (5.00 Ksi) for foundations, grade beams, columns.

Yield Strength of Steel

fy = 414 MPa (60 Ksi) for all reinforcing steel.

5. DESIGN LOADS

5.1. Dead Load(DL)

5.1.1. Roof Dead Loads

• Provision for Electrical Lighting	= 0.05 kN/m²
Ceiling	
Suspended steel channel system	= 0.10 kN/m²
Acoustical Fiber Tile (16mm)	= 0.05 kN/m²
Sub-total	= **0.2 kN/m²**

- Hollow Core Slab
  (HCS-250)(To be confirmed by Vendor) b/w Grid Line's (1 & 2) = **3.43 KN/m2**

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Prestressed Hollowcore Floor

Overall Structural Depth (mm)	Self Wt- KN/M2	No. of Cores in Unit	Spans indicated below allow for Characteristic Service Load (Live Load) Plus Self Weight Plus 1.5Kn/m² for Finishes Characteristic Service Loads Kn/m²							
			0.75	1.50	2.00	2.50	3.00	4.00	5.00	10.00
			Clear Span in metres							
100	2.38	0	5.39	5.19	5.08	4.91	4.72	4.40	4.14	3.30
150	2.52	11	7.40	7.40	7.40	7.40	7.28	6.80	6.40	5.12
150 (Dense	2.94	11	7.40	7.40	7.40	7.35	7.08	6.64	6.27	5.06
200	3.10	11	9.40	9.40	9.40	9.40	9.26	8.69	8.21	6.65
250	3.43	6	10.80	10.80	10.80	10.80	10.80	10.80	10.56	8.60
300	3.88	6	13.50	13.50	13.50	13.50	13.18	12.44	11.81	9.70
350	4.23	6	15.00	15.00	15.00	15.00	14.89	14.08	13.39	11.06
400	4.57	6	16.00	16.00	16.00	16.00	15.97	15.13	14.42	11.98
450	4.92	6	18.00	18.00	18.00	17.70	17.19	16.33	15.58	13.02
500	5.27	6	18.00	18.00	18.00	18.00	18.00	17.13	16.38	13.76

- Built-in Water Proofing System

100mm thk Polystyrene Foam

0.01 kN/m² per 25.4mm thickness x (100/25.4mm) = 0.039 kN/m²

Water proofing membrane

Bituminous, smooth surface = 0.07 kN/m²

90 mm thick average cement foam concrete screed Lightweight

Concrete 0.38 kN/m² per 25mm thickness x (90/25mm) = 1.368 kN/m²

75 mm Concrete screed above HCS = 21 x 0.075 =1.6 kN/m²

Sub-total = **3.1 kN/ m²**

Therefore

Total Dead Load b/w Grid's (1& 2) =3.43+0.2+3.1= 6.53 **kN/m²**

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TOTAL DL as UDL ON BEAMS

- UDL on Ledger Beams Along Grid Line-1 & 2 = 9.05 x 6.53/2 = **29.50 Kn/m**

Moment on all the External Beams due to Cantilever Portion of slab and Screed

= (0.15x24+ (0.05+0.2)/2x20)*0.6*0.6/2 = **1 knm/m**

5.1.2. Ground Floor Walls

- 250 tyke. Outer CMU (Partially Grouted Cell@800 c-c)

16 x 0.25 x (3.5-0.5) m clear height = **12 kN/m**

5.2. Live Load

5.2.1. Roof Live Loads(Lr)

Live loads are with reference to Clause-4.2.2, MD-216-1000-EG-ST-CRT-1001

Ordinary Flat (Inaccessible) = **0.5 kN/m²**

TOTAL Lr as UDL ON BEAMS

- UDL on Ledger Beams Along Grid Line-1 = 9.05 x 0.50/2 = **2.26 Kn/m**

Moment on all the External Beams due to Live Load on

Cantilever Portion of slab = 0.5*0.6*0.6/2 = **0.09 kN/m /m**

5.3. Wind Load(WL)

Wind loads were automatically generated in STAAD as per ASCE-07. The parameters used for the wind load generation are as follows (Refer STRUCTURAL DESIGN CRITERIA NON PROCESS BUILDINGS Doc No MD-216-0000-EG-ST-CRT-0001)

All structures and foundations (except for buried structures) designed and shall be constructed to resist the wind effects determined in accordance with Section-7.2, SBC-301-Analytical Procedure.

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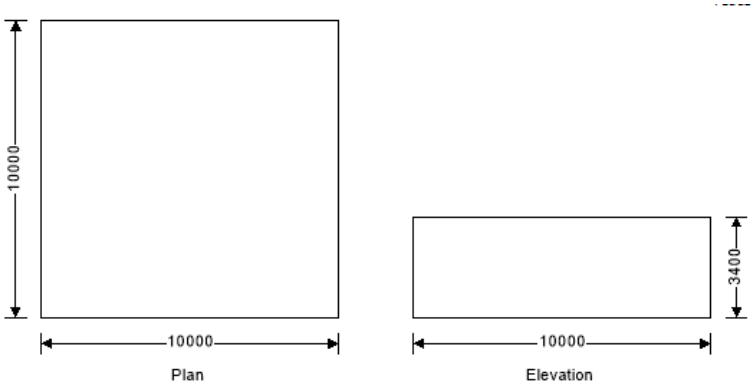
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5.4. Wind Load (Analytical Verification)

WIND LOADING (SBC-301)

Using the directional design method



Building data

Type of roof;	Flat
Length of building;	b = 10000 mm
Width of building;	d = 10000 mm
Height to eaves;	H = **3400** mm
Mean height;	h = **3400** mm

General wind load requirements

Basic wind speed;	V = **49** m/s (or) 177 kmph (Refer Clause-4.2.6, Doc- MD-216-1000-EG-ST- CRT-1001)
Risk category;	II(As per SBC-301, Table 1.6-1)
Velocity pressure exponent coeff ;	K_d = **0.85**(As per SBC-301, Table 6.4-1)
Exposure category;	D(As per SBC-301, Table 6.4.2.3)
Enclosure classification;	Partially enclosed buildings(As per SBC-301, Section-6.2)
Internal pressure coef +ve;	GC_{pi_p} = **0.55**(As per SBC-301, Figure-7.2-1)
Internal pressure coef –ve;	GC_{pi_n} = **-0.55**(As per SBC-301, Figure-7.2-1)
Gust effect factor;	G_f = **0.85**(As per SBC-301, Section-7.2.7.1)
Importance Factor;	1(As per SBC-301, Section-9.5)

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Topography

Topography factor not significant; K_{zt} = 1.0(As per SBC-301, Section-6.4.3.2)

Velocity pressure equation; $q = 0.0473x10^{-3}Kn/m^2 \times K_z \times K_{zt} \times K_d \times V^2 \times I$

V in Km/hr(As per SBC-301, Section-7.2.8)

Velocity pressures table

z (mm)	K_z (SBC-301,Table 7.2-2)	q_z (kN/m^2)
3400	1.03	1.29
3401	1.03	1.29

Pressures and forces

Net pressure; $p = q \times G_f \times C_{pe} - q_i \times GC_{pi}$;

Net force; $F_w = p \times A_{ref}$;

Walls load case 1 - Wind 0, GC_{pi} 0.55, - c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.40	0.80	1.29	0.17	33.99	5.69
B	3.40	-0.50	1.29	-1.26	34.00	-42.72
C	3.40	-0.70	1.29	-1.48	34.00	-50.16
D	3.40	-0.70	1.29	-1.48	34.00	-50.16

Walls load case 2 - Wind 0, GC_{pi} -0.55, + c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.40	0.80	1.29	1.58	33.99	53.86
B	3.40	-0.50	1.29	0.16	34.00	5.48

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Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
C	3.40	-0.70	1.29	-0.06	34.00	-1.97
D	3.40	-0.70	1.29	-0.06	34.00	-1.97

Walls load case 3 - Wind 90, GC_{pi} 0.55, - c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.40	0.80	1.29	0.17	33.99	5.69
B	3.40	-0.50	1.29	-1.26	34.00	-42.72
C	3.40	-0.70	1.29	-1.48	34.00	-50.16
D	3.40	-0.70	1.29	-1.48	34.00	-50.16

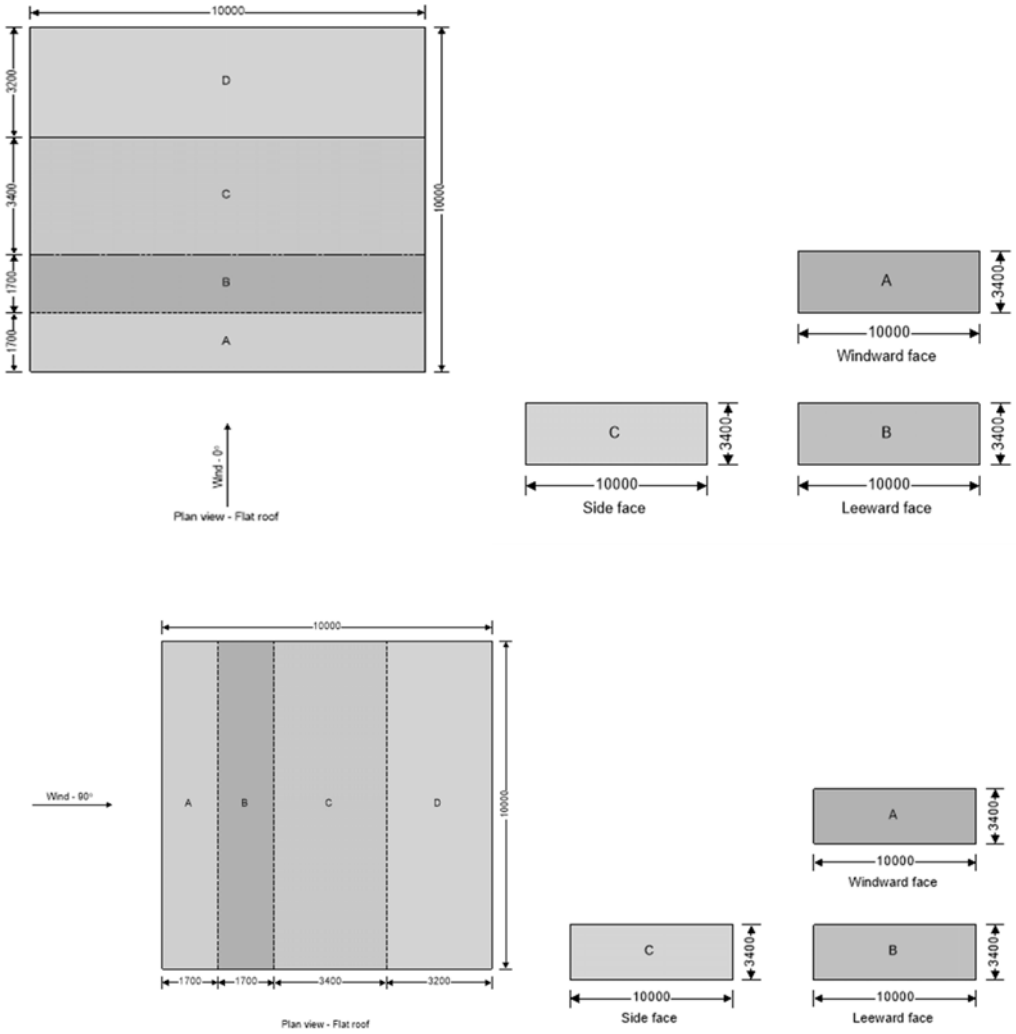
Walls load case 4 - Wind 90, GC_{pi} -0.55, + c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.40	0.80	1.29	1.58	33.99	53.86
B	3.40	-0.50	1.29	0.16	34.00	5.48
C	3.40	-0.70	1.29	-0.06	34.00	-1.97
D	3.40	-0.70	1.29	-0.06	34.00	-1.97

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So from the above analysis, the Wind Loads generated in STAAD are in accordance with SBC-301-2007-Structural Loading & Forces.

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5.5. Seismic Load(E)

Earthquake loads were automatically generated in STAAD as per IBC 2003. The basic parameters affecting the generation of seismic loads is shown below:

Importance Factor, I = 1.0(As per SBC-301, Section-9.5)

Design Spectral Response Acceleration (Refer Clause-4.2.7b, MD-216-1000-EG-ST- CRT-1001)

Ss = 0.1g (Spectral Response Acceleration 0.2 second)

S1 = 0.02g (Spectral Response Acceleration 1 second)

S_{D_1} = 2/3*Fv*S1 = 2/3 * 2.4 * 0.02 = 0.032 (As per SBC-301, Section-9.4.4)

S_{DS} = 2/3*Fa*Ss = 2/3 * 1.6 * 0.1 = 0.106 (As per SBC-301, Section-9.4.4)

Where S_{D_1} Damped Design Spectral Response Acceleration@1 Sec

S_{DS} Damped Design Spectral Response Acceleration@0.2 Sec

Site Class = D

Site Coefficient, Fa = 1.60 (As per SBC-301, Table 9.4.3a)

Site Coefficient, Fv = 2.40 (As per SBC-301, Table 9.4.3b)

Building period Coefficient, Ct=0.04 (As per SBC-301, Table 10.9.3.2)

For Ordinary reinforced concrete moment frame,

Response Modification Factor, Rx = 2.5 (As per SBC-301, Table 10.9.3.2)

Response Modification Factor, Rz = 2.5 (As per SBC-301, Table 10.2)

System Over Strength Factor, Omega = 3 (As per SBC-301, Table 10.2)

Deflection Amplification factor, Cd = 2.5 (As per SBC-301, Table 10.2)

5.6. Thermal/Temperature Loads (Self-restraining and Thermal Expansion and Contraction Loads)

An ambient temperature ranges from a maximum of 50° C to a low of 0° C. In direct exposure to sunlight, the temperature shall be assumed to rise a minimum of 15° C above ambient. As a minimum, the temperature gradient for thermal loads shall be +25° C and -15° C with the most severe condition governing the design.

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6. DESIGN COMBINATIONS

Service Loads (Allowable Stress Design) (SBC-301, Section 2.4.1)

D

D + T

D +Lr

D +0.75Lr

D + (W or 0.7E)

D + 0.75W + 0.75Lr

0.6D + W

0.6D + 0.7E

Factored loads (Strength Design) (SBC-301, Section 2.3.2)

1.4D

1.4D+1.4T

1.2(D + T) + 0.5Lr

1.2D + 1.6Lr + 0.8W

1.2D + 1.6W + 0.5Lr

1.2D + 1.0E

0.9D + 1.6W

0.9D + 1.0E

Where:

D = Dead Load

Lr = Live Load

W = Wind Load

E = Seismic Load

T = Temperature Load

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7. CRACKED SECTIONS

In lieu of a more accurate analysis of the effects of cracking of concrete, provisions of SBC-304-Section 10.11 was used.

For Beams:	I = 0.35Ig
For Columns:	I = 0.70Ig
For Walls:	
Un-cracked	I = 0.70Ig
Cracked	I = 0.35Ig
For Slabs	I = 0.25Ig

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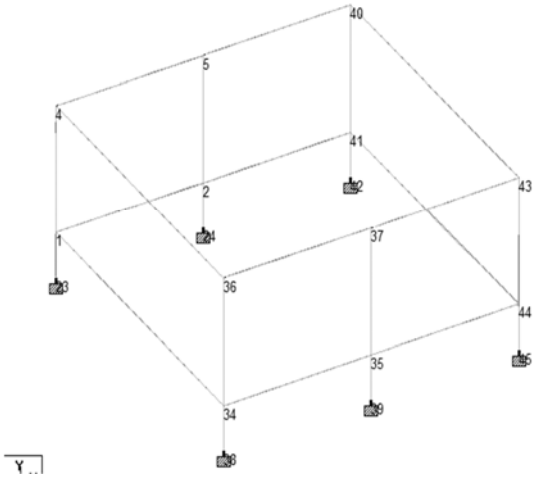
8. DEFLECTIONS

8.1. Maximum Allowable Deflection

L/240: for member load with live load plus dead load

L/600: for members supporting masonry under total loads

Where, L = span of member between support



All / Summary /

ABLUTION BUILDING.std - Node Displacement

Node	L/C	X-Trans mm	Y-Trans mm	Z-Trans mm	Absolute mm	X-Rotan rad	Y-Rotan rad	Z-Rotan rad
5	17	0.099	-0.179	10.560	10.562	0.003	0.000	-0.000
5	21	0.164	-0.298	10.521	10.526	0.002	0.000	-0.000
37	21	0.161	-0.298	9.220	9.226	0.003	0.000	-0.000
37	17	0.096	-0.179	9.181	9.183	0.003	0.000	-0.000
5	15	0.172	-0.311	7.816	7.824	0.002	0.000	-0.000
37	15	0.170	-0.312	6.989	6.998	0.002	0.000	-0.000
37	20	1.553	-0.298	6.658	6.843	0.002	0.000	-0.000
5	20	1.553	-0.298	-6.658	6.843	-0.002	-0.000	-0.000
5	16	1.488	-0.179	-6.619	6.786	-0.002	-0.000	-0.000
37	16	1.488	-0.179	6.619	6.786	0.002	0.000	-0.000
4	21	0.174	-0.179	5.722	5.728	0.001	-0.001	-0.001
40	21	0.153	-0.184	5.714	5.719	0.001	0.001	0.001
4	17	0.098	-0.104	5.713	5.715	0.001	-0.001	-0.000
40	17	0.100	-0.108	5.704	5.706	0.001	0.001	0.000
36	17	0.093	-0.119	5.681	5.683	0.001	-0.001	-0.000
43	17	0.099	-0.122	5.672	5.674	0.001	0.000	0.000
		2.225	0.000	-0.000	0.099	0.099		
		3.337	-0.000	-0.000	0.153	0.153		
		4.450	0.000	0.000	0.000	0.000		
	2 EQK LOAD	0.000	0.000	0.000	0.000	0.000		
		1.112	-0.000	-0.064	-0.005	0.065		
		2.225	-0.000	-0.026	-0.006	0.026		
		3.337	-0.000	0.026	-0.003	0.026		
		4.450	0.000	0.000	0.000	0.000		

Maximum Deflection Attained is 11mm<<<33mm (H/100) .Hence Structure is safe against Drift Check.

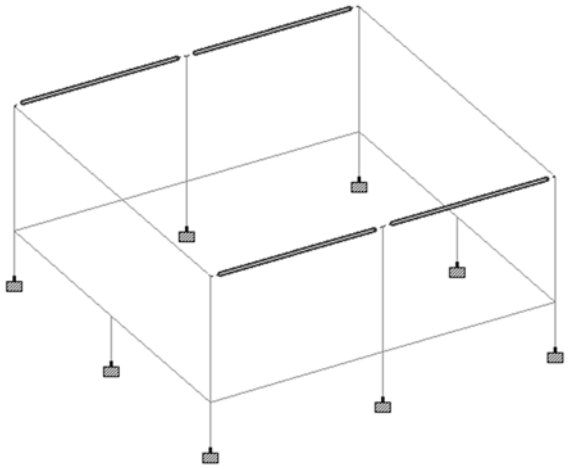
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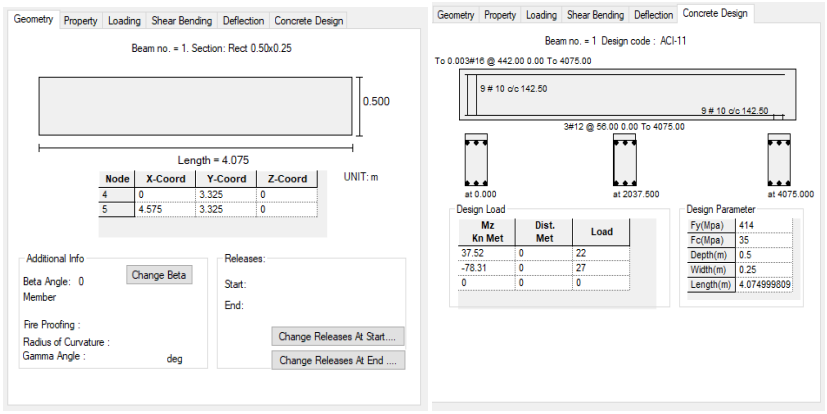
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9. CONCRETE ROOF BEAM DESIGN

9.1. Design of Ledger Beam(LB1)





Design of Ledger Beam (LB1)

Maximum Moment required from Analysis = 78 **Knm (Hogging)**

Therefore Ast.req =

$$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

Where b=250mm, Fy=414 Mpa, d=500-40-10-20/2=440mm, Fc'=35 Mpa & ϕ =0.9

Ast1.req = **476 mm²**

Pl

$$A_{s,min} = \frac{0.25\sqrt{f_c'}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

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Ast.min=
(As per clause-10.5-ACI-318)

Ast.min= **406mm²**

Maximum Moment required from Analysis = **37 Knm (Sagging)**

Ast2.req = **240 mm²**

Hence Provide 3-16 dia-Full Straight Bars Top & 3-16 dia-Full Straight Bottom

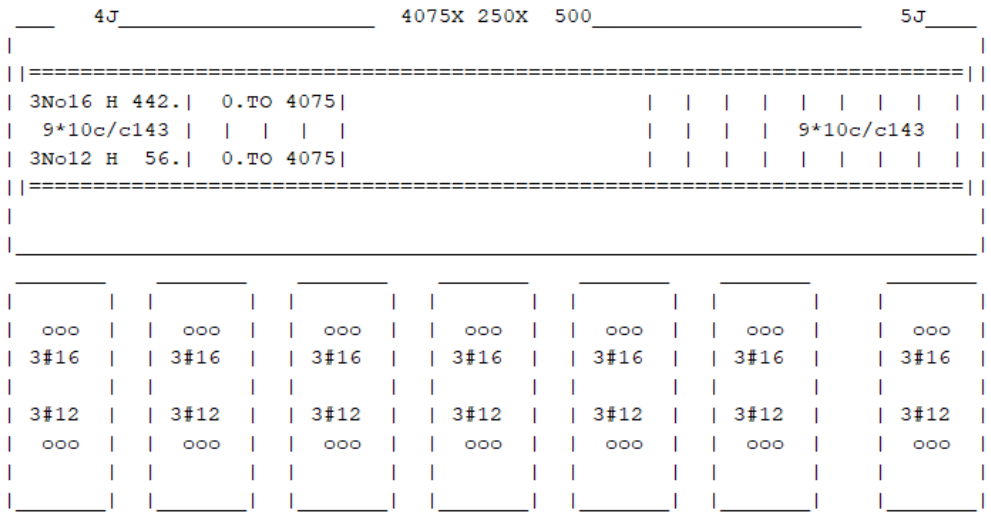
Ast1.pro= **602mm² >>>>>>(Ast.min, Ast1.req)**

Ast2.pro= **602mm² >>>>>>(Ast.min, Ast2.req)**

9.1.1. Staad Output For LB1

```
AT START SUPPORT - Vu=   29.74 KNS  Vc=  107.30 KNS  Vs=    0.00 KNS
Tu=    7.11 KN-MET  Tc=    4.1 KN-MET  Ts=    9.5 KN-MET  LOAD    27
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 1098. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   2.34 SQ.CM.

AT END   SUPPORT - Vu=   37.74 KNS  Vc=  108.58 KNS  Vs=    0.00 KNS
Tu=    5.20 KN-MET  Tc=    3.8 KN-MET  Ts=    6.9 KN-MET  LOAD    32
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 1098. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   1.71 SQ.CM.
```

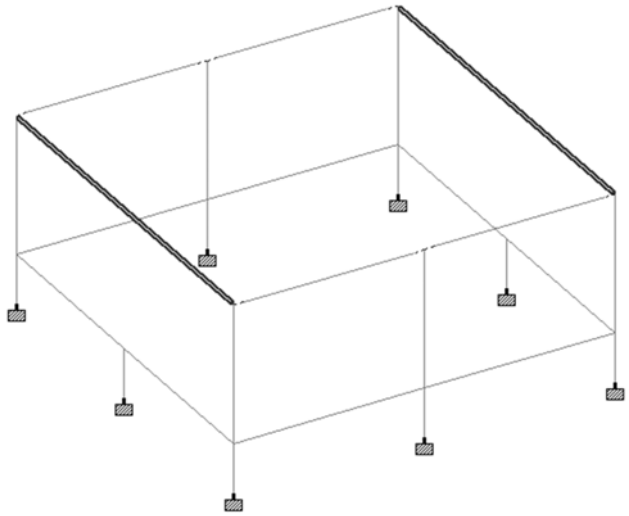


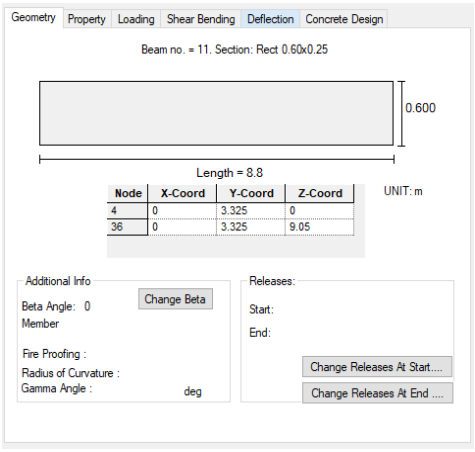
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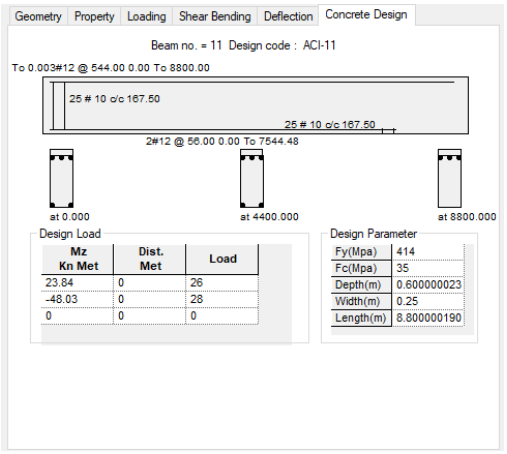
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9.2. Design of Beam-(B1)







Provide minimum 3-16 dia-Full Straight Bars Top & Bottom

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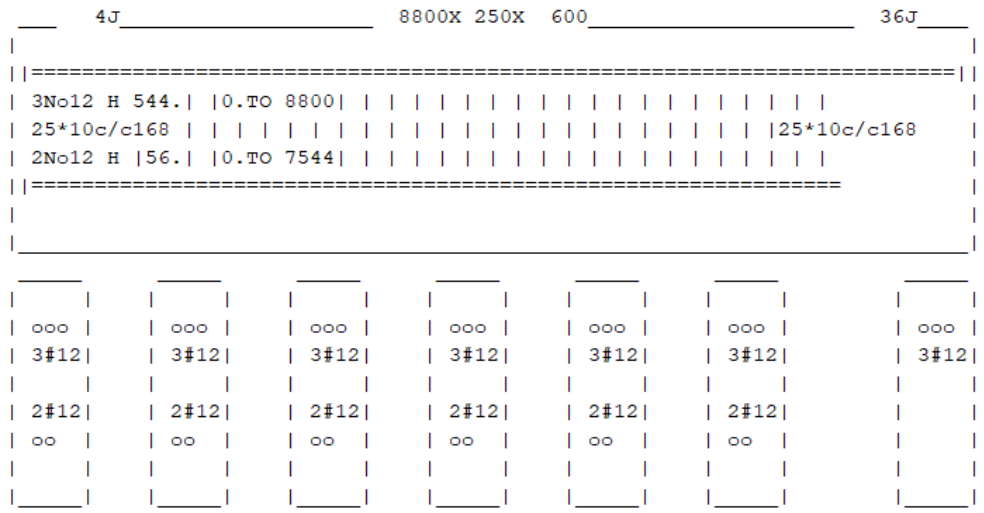
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9.2.1. Staad Output For B1

```
                B E A M  N O.    11 D E S I G N  R E S U L T S - SHEAR


   AT START SUPPORT - Vu=   19.14 KNS  Vc=  129.54 KNS  Vs=    0.00 KNS
   Tu=    5.60 KN-MET  Tc=    5.0 KN-MET  Ts=    7.5 KN-MET  LOAD    23
   STIRRUPS ARE REQUIRED FOR TORSION.
   REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
   PROVIDE 10 MM 2-LEGGED STIRRUPS AT 168. MM  C/C FOR 3860. MM
   ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   1.74 SQ.CM.

   AT END   SUPPORT - Vu=   19.14 KNS  Vc=  129.54 KNS  Vs=    0.00 KNS
   Tu=    5.60 KN-MET  Tc=    5.0 KN-MET  Ts=    7.5 KN-MET  LOAD    23
   STIRRUPS ARE REQUIRED FOR TORSION.
   REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
   PROVIDE 10 MM 2-LEGGED STIRRUPS AT 168. MM  C/C FOR 3860. MM
   ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   1.74 SQ.CM.
```



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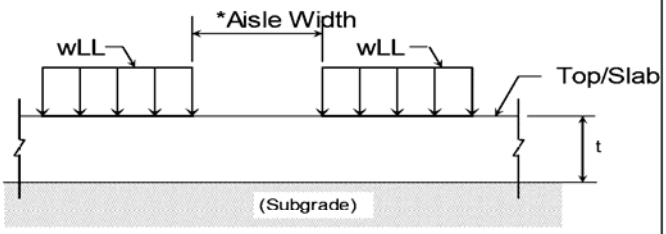
10. CONCRETE GRADE SLAB DESIGN

CONCRETE SLAB ON GRADE ANALYSIS

For Slab Subjected to Stationary Uniformly Distributed Live Loads

Input Data:

Slab Thickness, t =	3.937	in.	100.000	mm		
Concrete Strength, f 'c =	5000	psi	35.000	kpa		
Subgrade Modulus, k =	39.7866	pci	10800	Kn/m3	(AS PER SECTION-7.7-SOIL REPORT)	
Factor of Safety, FS =	1.500		1.500			
Uniform Live Load, wLL =	104.45	psf	5.00	kpa		



Concrete Slab on Grade with Uniform Loads

Results:

Design Parameters:

Modulus of Rupture, MR =	636.40	psi	4387.79	kpa	MR = 9*SQRT(f 'c)
Allow. Bending Stress, Fb =	424.26	psi	2925.19	kpa	Fb = MR/FS
Modulus of Elasticity, Ec =	4030509	psi	27789329	kpa	Ec = 57000*SQRT(f 'c)
Poisson's Ratio, μ =	0.15		0.15		μ = 0.15 (assumed for concrete)
Radius of Stiffness, Lr =	26.94	in.	684.37	mm	Lr = (Ec*t^3/(12*(1-μ^2)*k))^0.25
Critical Aisle Width, Wcr =	4.96	ft.	1511.77	mm	Wcr = (2.209*Lr)/12

Stationary Uniformly Distributed Live Loads:

wLL(allow) =	682.05	psf	wLL(allow) = 257.876*Fb*SQRT(k*t/Ec)
	32.65	kpa	**wLL(allow) >= wLL, O.K.**

Provide Minimum Reinforcement of 0.14%@both faces = 0.14%(h) = 140 mm2/M

Table 24.4.3.2—Minimum ratios of deformed shrinkage and temperature reinforcement area to gross concrete area

Reinforcement type	f_y, psi	Minimum reinforcement ratio	
Deformed bars	< 60,000	0.0020	
Deformed bars or welded wire reinforcement	≥ 60,000	Greater of:	$\frac{0.0018 \times 60,000}{f_y}$
			0.0014

Provide WWF-8mm Dia@200 c-c Both Ways-TOP LAYER

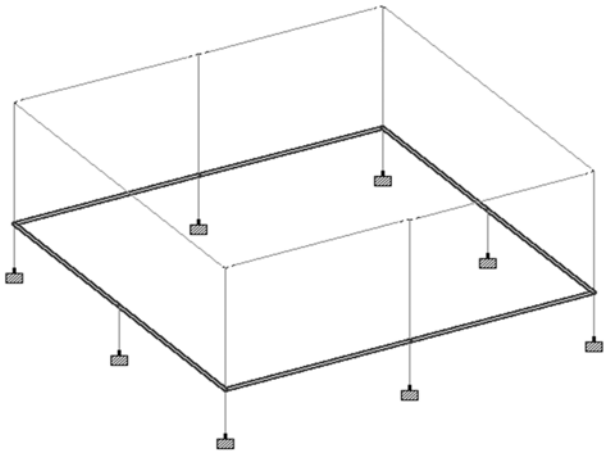
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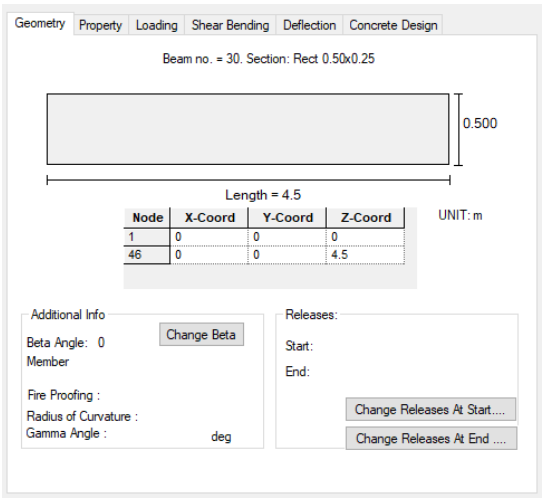
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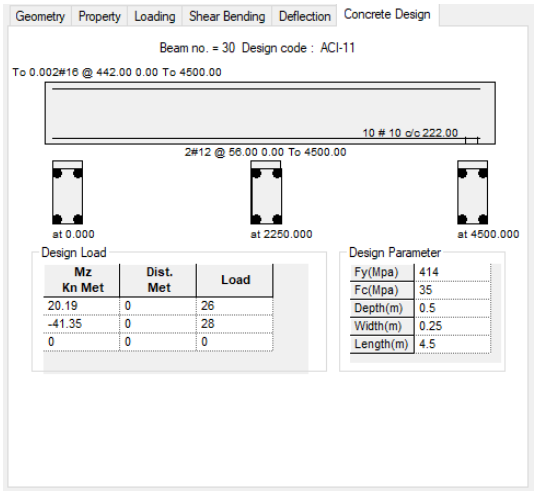
11. DESIGN OF GRADE BEAM

11.1. Grade Beam(GB1)



MOMENT ENVELOPE FOR BEAMS GB1





Provide minimum 3-16 dia-Full Straight Bars Top & 3-16 dia-Full Straight Bars Bottom

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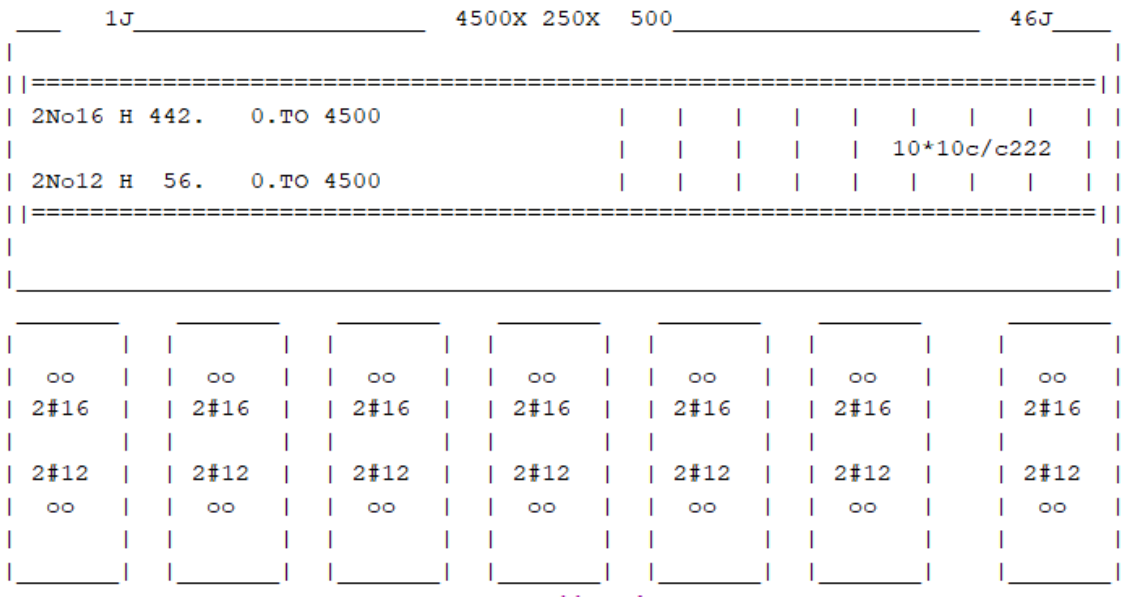
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11.1.1. STAAD OUTPUT FOR GB1

```
                B E A M  N O.    30 D E S I G N  R E S U L T S - SHEAR


  AT START SUPPORT - Vu=   25.29 KNS  Vc=  108.39 KNS  Vs=    0.00 KNS
  Tu=    0.04 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD     3
                    STIRRUPS ARE NOT REQUIRED.
  AT END   SUPPORT - Vu=   40.37 KNS  Vc=  106.38 KNS  Vs=    0.00 KNS
  Tu=    0.05 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD    22
  NO STIRRUPS ARE REQUIRED FOR TORSION.
  REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
  PROVIDE 10 MM 2-LEGGED STIRRUPS AT 222. MM  C/C FOR 1810. MM
```

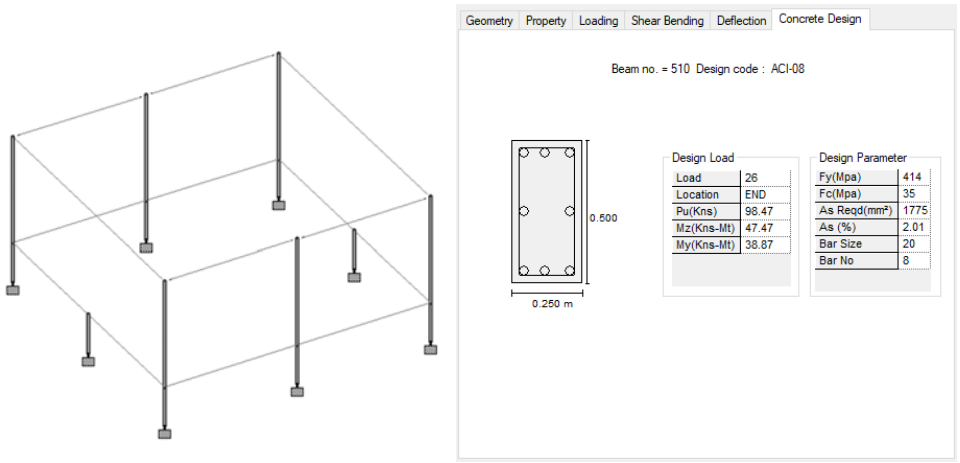


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12. CONCRETE COLUMN DESIGN



Column's C1

However provide equivalent 8-Dia20 Vertical Reinforcement in Column C1.

12.1. Staad Output For Column C1

```
        COLUMN   NO.    510   DESIGN PER ACI 318-08 - AXIAL + BENDING


  FY - 413.7   FC -   35.0 MPA,   RECT SIZE - 250.0 X 500.0 MMS, TIED
          AREA OF STEEL REQUIRED =   1775.0   SQ. MM

  BAR CONFIGURATION        REINF PCT.    LOAD    LOCATION    PHI
  ------------------------------------------------------------

--------------------------------< PAGE 25 Ends Here >----------------------------
   STAAD SPACE                                                 -- PAGE NO.    26

    8 - 20 MM                2.011        26       END      0.650
  (PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
  TIE BAR NUMBER    10 SPACING 250.00 MM
```


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13. CANTILEVER SLAB DESIGN

DESIGN OF CANTILEVER SLAB

Geometrical Properties			Material Properties			
Span to center of support "sp_1"	0.8	met	Weight of reinforced concrete "γ_r	24		kN/m³
Clear Span	0.6	met	Strength of concrete "f'_c"	35		N/mm²
Clear span + depth of slab "sp_2"	0.75	met	Strength of steel "f_y"	420		N/mm²
Max. eff span le = min (sp_1;sp_2)	0.8	met	Clear cover to concrete("c")	20		mm
Width of Slab "b"	1000	mm	Flexure strength reduction factor	0.9		
Thickness of slab "t"	150	mm	Shear strength reduction factor	0.75		
Eff. Depth "d"	125		Modulus of Elasticity of steel "Es"	2E+05		N/mm²
Loading			Modulus of Elasticity of concrete	27806		N/mm²
Self Weight of slab	3.6	kN/m²	Modular Ratio "n = Es/Ec"	7.19		
Floor Finish	3.5	kN/m²	**Provided Reinforcement**			
Wall load	0	kN/m²	Top layer Nos.	6	Φ 10	mm
Total dead Load "U_{dl}"	7.1	kN/m²	Bottom layer Nos.	0	Φ 12	mm
Live load	1	kN/m²	Area of Top steel "$A_{s\ t}$"		468	mm²
Total Ultimate Load "$Uu_{,tot}$"(1.2D+1.6L)	10.12	kN/m²	Area of Bottom steel "$A_{s\ b}$"		0	mm²
Moment due to Overhang Point Load(0.125*0.15*24*0.6)	0.27	kNm	"db' = 10 mm			
Calculations						
Shear force Vu = U*le	**8.096**	**kN**				
Neg. Moment Mu = U*le²/2	**3.51**	**kNm**				

Check for Section capacity

Depth of comp. block "a" = $A_{s\ t}*f_y/(0.85*f'_c*b)$	6.61	mm						
Section capacity ΦMn = Φ*$A_{s\ t}$*f_y*(d-a/2)	**21.94**	**kNm**	**O.K**	**>**	**3.51**	**kNm**	(Mu)	

Shear requirements

Shear capacity ΦVc = Φ*0.17*√f'c*b*d	**94.29**	**kN**	**O.K**	**>**	**8.096**	**kN**	(Vu)	

Serviceability requirements

d' = c+d_b/2	26	mm
B = b/(n*$A_{s\ t}$)	0.297	
r = (n-1)*$A_{s\ b}$/(n*$A_{s\ t}$)	0	
k_d = (√{2*d*B(1+r*d'/d)+(1+r)²}-(1+r))/B	29.01	mm
I_{cr} = bkd³/3+n*$A_{s\ t}$*(d-k_d)²+(n-1)*$A_{s\ b}$*(k_d-d'	3.915E+07	mm⁴
Calculate Gross moment of inertia		
Area of slab A = t*b	2E+05	mm²
Depth of neutral axis "y"	75	mm
Distance of top fiber from N.A "y_t"	75	mm
Gross moment of inertia "I_g"	2.813E+08	mm⁴
Allowable stress fr = 0.62√f'c	3.67	mm²
Cracking moment M_{cr} = fr*I_g/y_t	13.75	kNm

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Service moments

Moment due to dead load "M_{dl}"	2.54	kNm	
Moment due to live load "Mll"	0.32	kNm	
Total Moment Ms = $M_{dl}+M_{ll}$	2.86	kNm	
Deflection due to dead + live loads			
R(dl+ll) = M_{cr}/Ms	1.000		
I_e(dl+ll) = $R^3*I_g+(1-R^3)*I_{cr}$	2.813E+08	mm^4	
δ = K*5/48*Ms*le^2/(Ec*I_e)	0.08	mm	**K = 3.2 (conservative)**
Deflection due to dead load			
R(dl) = M_{cr}/Md	5.411		
I_e(dl) = $R^3*I_g+(1-R^3)*I_{cr}$	3.839E+10	mm^4	
δ = K*5/48*Md*le^2/(Ec*I_e)	0.00	mm	
Deflection due to live load			
Live load deflection = dl+ll deflection - dl deflection	0.08	mm	
Allowable immediate deflection due to Live load = per ACI 318 Tab 9.5b	**I_e/360 2.22 mm > 0.08 mm O.K**		
Long term deflection			
% of liveload sustained "SL"	50	%	
M_{sus} = M_{dl}+SL*M_{ll}	2.70	kNm	
R(sus) = M_{cr}/Msus	5.091		
I_e(sus) = $R^3*I_g+(1-R^3)*I_{cr}$	3.198E+10		
δ = K*5/48*Msus*le^2/(Ec*I_e)	0.00	mm	
Magnification factor for creep and shrinkage			
λ_Δ = ξ / (1+50*ρ')	2.000		
Where:-			
ξ = 2 for 5 years or more	cl.9.5.2.5 ACI 318		
ρ' = $A_{s\ b}$/(b*d)	0		
Total deflection = λ_Δ*sus deflection + ll deflection	0.079	mm	
Allowable deflection = I_e/240	**3.333 mm > 0.079 mm O.K**		

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14. FOUNDATION DESIGN

Foundations have been designed based on the Soil Report Recommendations. The minimum Embedment Depth of Foundations has been considered as 1.2m & applicable SBC of 500 Kpa. (Refer Doc-MD-216-3000-EG-CV-CAL-1008-A01 for Details)

TABLE 4: RECOMMENDED NET ALLOWABLE BEARING CAPACITIES

PRPJECT: MANSOURAH - MANSSARAH GOLD PROJECT
CLIENT: L & T CONSTRUCTION

Profile / Borehole	Foundation Type	Depth of Foundation below FGL	Footing Size (B x L)	Net Allowable Bearing Capacity Based on		Recommended Net Allowable Bearing Capacity	Estimated Total Settlement
				Settlement	Shear Failure		
-	-	(m)	(m)	(kN/m^2)	(kN/m^2)	(kN/m^2)	(mm)
BH-01	Square Isolated	1.20	1.0 x 1.0	1500	570	570	25
		1.50		2000	850	850	25
		1.20	2.0 x 2.0	1000	750	750	25
		1.50		1200	980	980	25
		1.20	3.0 x 3.0	775	900	775	25
		1.50		950	1120	950	25
		1.20	4.0 x 4.0	625	1040	625	25
		1.50		750	1250	750	25
		1.20	5.0 x 5.0	525	1172	525	25
		1.50		600	1342	600	25

Refer Attachment- II for Foundation Design calculations

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REFERENCES

1.	BUILDING PLANS	MD-216-4000-EG-AR-GAD-3025
2.	BUILDING ELEVATIONS & SECTIONS	MD-216-4000-EG-AR-GAD-3026
3.	LAYOUT & DETAILS OF FOUNDATION, COLUMN & GRADE BEAM	MD-216-4000-EG-CV-GAD-1071
4.	LAYOUT & DETAILS OF ROOF & GRADE SLAB	MD-216-4000-EG-CV-DED-1072
5.	GEOTECHNICAL REPORT	MD-216-3000-EG-CV-CAL-1008
6.	STRUCTURAL DESIGN CRITERIA	MD-216-0000-EG-ST-CRT-0001

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ATTACHMENT 1 STAAD INPUT DATA

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```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 18-Oct-19
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 0 0; 2 4.575 0 0; 4 0 3.325 0; 5 4.575 3.325 0; 23 0 -1.2 0;
24 4.575 -1.2 0; 34 0 0 9.05; 35 4.575 0 9.05; 36 0 3.325 9.05;
37 4.575 3.325 9.05; 38 0 -1.2 9.05; 39 4.575 -1.2 9.05; 40 9.15 3.325 0;
41 9.15 0 0; 42 9.15 -1.2 0; 43 9.15 3.325 9.05; 44 9.15 0 9.05;
45 9.15 -1.2 9.05; 46 0 0 4.5; 47 0 -1.2 4.5; 48 9.15 0 4.5; 49 9.15 -1.2 4.5;
MEMBER INCIDENCES
1 4 5; 3 1 4; 4 2 5; 11 4 36; 25 1 2; 30 1 46; 31 1 23; 32 2 24; 497 36 37;
498 34 36; 499 35 37; 500 35 34; 501 34 38; 502 35 39; 505 5 40; 506 41 40;
507 2 41; 508 41 42; 509 37 43; 510 44 43; 511 44 35; 512 44 45; 513 40 43;
514 41 48; 515 46 34; 516 46 47; 517 48 44; 518 48 49;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17184e+007
POISSON 0.17
DENSITY 23.6158
ALPHA 5e-006
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27578.9
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
3 4 498 499 506 510 PRIS YD 0.5 ZD 0.25
1 25 30 497 500 505 507 509 511 514 515 517 PRIS YD 0.5 ZD 0.25
MEMBER PROPERTY AMERICAN
11 513 PRIS YD 0.6 ZD 0.25
MEMBER PROPERTY
31 32 501 502 508 512 516 518 PRIS YD 0.5 ZD 0.25
CONSTANTS
BETA 90 MEMB 516 518
MATERIAL CONCRETE ALL
SUPPORTS
23 24 38 39 42 45 47 49 FIXED
******************************************************************************
MEMBER OFFSET
1 497 505 509 START 0.25 0 0
1 497 505 509 END -0.25 0 0
11 513 START 0 0 0.125
11 513 END 0 0 -0.125
MEMBER CRACKED
1 11 25 30 497 500 505 507 509 511 513 -
514 REDUCTION RIX 0.350000 RIY 0.350000 RIZ 0.350000
3 4 31 32 498 499 501 502 506 508 510 512 516 -
518 REDUCTION RIX 0.700000 RIY 0.700000 RIZ 0.700000
DEFINE IBC 2003
SDS 0.1 SD1 0.03 S1 0.02 IE 1 RX 2.5 RZ 2.5 SCLASS 2
SELFWEIGHT 1
MEMBER WEIGHT
25 30 500 507 511 514 515 517 UNI 12
1 497 505 509 UNI 29.5
DEFINE WIND LOAD
TYPE 1 WIND X WINDWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
```

```
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 3.500 M 10.000 M 10.000 M 2.000 0.010 0 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.893 0.800 -0.550
!> END GENERATED DATA BLOCK
INT 1.64157 1.64157 HEIG  0  4.572
TYPE 2 WIND X LEEWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 3.500 M 10.000 M 10.000 M 2.000 0.010 1 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.893 -0.500 0.550
!> END GENERATED DATA BLOCK
INT -1.29377 -1.29377 HEIG  0  3.5
TYPE 3 SIDEWARD X
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 3.500 M 10.000 M 10.000 M 2.000 0.010 2 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.893 -0.700 0.550
!> END GENERATED DATA BLOCK
INT -1.52563 -1.52563 HEIG  0  3.5
TYPE 4 WIND Z WINDWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 3.500 M 10.000 M 10.000 M 2.000 0.010 0 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.893 0.800 -0.550
!> END GENERATED DATA BLOCK
INT 1.64157 1.64157 HEIG  0  4.572
TYPE 5 WIND Z LEEWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 3.500 M 10.000 M 10.000 M 2.000 0.010 1 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.893 -0.500 0.550
!> END GENERATED DATA BLOCK
INT -1.29377 -1.29377 HEIG  0  3.5
TYPE 6 SIDEWARD Z
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 3.500 M 10.000 M 10.000 M 2.000 0.010 2 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.893 -0.700 0.550
!> END GENERATED DATA BLOCK
INT -1.52563 -1.52563 HEIG  0  3.5
LOAD 1 LOADTYPE Seismic  TITLE EQK LOAD Z
IBC LOAD Z 1
PERFORM ANALYSIS
CHANGE
LOAD 2 LOADTYPE Seismic  TITLE EQK LOAD X
IBC LOAD X 1
PERFORM ANALYSIS
CHANGE
LOAD 3 LOADTYPE Dead  TITLE DL
SELFWEIGHT Y -1 LIST ALL
MEMBER LOAD
25 30 500 507 511 514 515 517 UNI GY -12
1 497 505 509 UNI GY -29.5
```

11 UMOM GZ 1
513 UMOM GZ -1
1 505 UMOM GX -1
497 509 UMOM GX 1
LOAD 4 LOADTYPE Live TITLE LL
MEMBER LOAD
1 497 505 509 UNI GY -2.26
11 UMOM GZ 0.1
513 UMOM GZ -0.1
1 505 UMOM GX -0.1
497 509 UMOM GX 0.1
LOAD 5 LOADTYPE Wind TITLE WL+X
WIND LOAD X 1 TYPE 1 YR 0 3.5
WIND LOAD -X 1 TYPE 2 YR 0 3.5
WIND LOAD -Z 1 TYPE 3 YR 0 3.5
WIND LOAD -Z -1 TYPE 3 YR 0 3.5
LOAD 6 LOADTYPE Wind TITLE WL+Z
WIND LOAD Z 1 TYPE 4 YR 0 3.5
WIND LOAD -Z 1 TYPE 5 YR 0 3.5
WIND LOAD -X 1 TYPE 6 YR 0 3.5
WIND LOAD -X -1 TYPE 6 YR 0 3.5
LOAD 7 LOADTYPE Temperature TITLE TEMP
TEMPERATURE LOAD
1 3 4 497 TO 499 505 506 509 510 513 TEMP 25
LOAD COMB 8 SLS: (DL)
3 1.0
LOAD COMB 9 SLS: (DL+T)
3 1.0 7 1.0
LOAD COMB 10 SLS: (D+LR)
3 1.0 4 1.0
LOAD COMB 11 SLS: (D+0.75LR)
3 1.0 4 0.75
LOAD COMB 12 SLS: (D+0.7EZ)
3 1.0 1 0.7
LOAD COMB 13 SLS: (D+0.7EX)
3 1.0 2 0.7
LOAD COMB 14 SLS: (D+0.75WX+0.75LR)
3 1.0 5 0.75 4 0.75
LOAD COMB 15 SLS: (D+0.75WZ+0.75LR)
3 1.0 6 0.75 4 0.75
LOAD COMB 16 SLS: (0.6D+WX)
3 0.6 5 1.0
LOAD COMB 17 SLS: (0.6D+WZ)
3 0.6 6 1.0
LOAD COMB 18 SLS: (0.6D+0.7EX)
3 0.6 2 0.7
LOAD COMB 19 SLS: (0.6D+0.7EZ)
3 0.6 1 0.7
LOAD COMB 20 SLS: (D+WX)
3 1.0 5 1.0
LOAD COMB 21 SLS: (D+WZ)
3 1.0 6 1.0
LOAD COMB 22 ULS: (1.4 DL)
3 1.4
LOAD COMB 23 ULS: (1.4D+1.4T+0.5LR)
3 1.4 4 0.5 7 1.4
LOAD COMB 24 ULS: (1.4D+1.4T+0.5LR)
3 1.4 4 0.5 7 1.4
LOAD COMB 25 ULS: (1.2D+0.80WX+1.6LR)

3 1.2 5 0.8 4 1.6
LOAD COMB 26 ULS: (1.2D+1.6WZ+0.5LR)
3 1.2 6 1.6 4 0.5
LOAD COMB 27 ULS: (1.2D+1.6WX+0.5LR)
3 1.2 4 0.5 5 1.6
LOAD COMB 28 ULS: (1.2D+1.6WZ+0.5LR)
3 1.2 4 0.5 6 1.6
LOAD COMB 29 ULS: (1.2D+EX)
3 1.2 2 1.0
LOAD COMB 30 ULS: (1.2D+EZ)
3 1.2 1 1.0
LOAD COMB 31 ULS: (0.90D+1.6WX)
3 0.9 5 1.6
LOAD COMB 32 ULS: (0.90D+1.6WZ)
3 0.9 6 1.6
LOAD COMB 33 ULS: (0.90D+EZ)
3 0.9 1 1.0
LOAD COMB 34 ULS: (0.90D+EX)
3 0.9 2 1.0
LOAD COMB 35 SLS: (DL+W-Z)
3 1.0
LOAD COMB 36 SLS: 0.6DL+WZ-
3 0.6
LOAD COMB 37 ULS: (0.9DL+1.6WZ-VE)
3 0.9
PERFORM ANALYSIS
START CONCRETE DESIGN
CODE ACI
FC 35000 MEMB 1 3 4 11 25 30 TO 32 497 TO 502 505 TO 514
CLB 0.04 MEMB 1 3 4 11 25 30 TO 32 497 TO 502 505 TO 514
CLS 0.04 ALL
CLT 0.04 MEMB 1 3 4 11 25 30 TO 32 497 TO 502 505 TO 514
MINSEC 10 ALL
SFACE 0.5 MEMB 1 497 505 509
EFACE 0.5 MEMB 1 497 505 509
DESIGN BEAM 1 11 25 30 497 500 505 507 509 511 513 TO 515 517
DESIGN COLUMN 3 4 31 32 498 499 501 502 506 508 510 512 516 518
END CONCRETE DESIGN
FINISH

ATTACHMENT 2 FOUNDATION DESIGN CALCULATIONS

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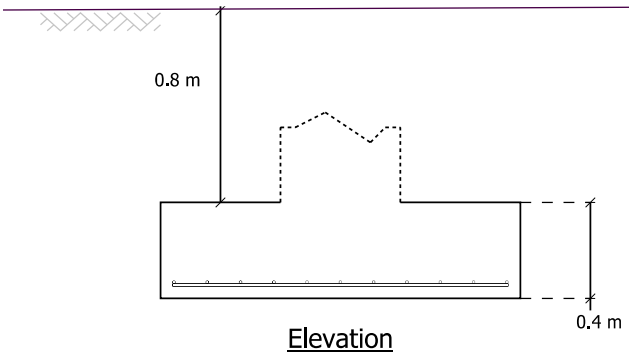
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Isolated Footing Design(ACI 318-08)

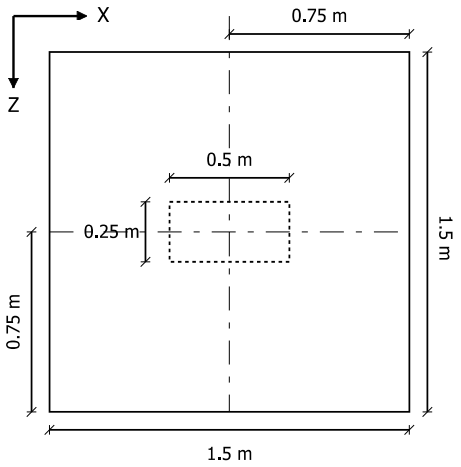
Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
24	1	1.500m	1.500m	0.400m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
24	11 - ϕ12	11 - ϕ12	N/A	N/A	N/A	N/A

FOUNDATION F1-GRID-1 & B



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 1500.000mm
Footing Width - Z (Fw) : 1500.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.500m

Column Width - Z (B_{col}) : 0.250m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 125.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.530
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.500m

Initial Width (W_o) = 1.500m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
8	SLS: (DL)	1.00	1.00	1.00
9	SLS: (DL+T)	1.00	1.00	1.00
10	SLS: (D+LR)	1.00	1.00	1.00
11	SLS: (D+0.75LR)	1.00	1.00	1.00
12	SLS: (D+0.7EZ)	1.00	1.00	1.00
13	SLS: (D+0.7EX)	1.00	1.00	1.00
14	SLS: (D+0.75WX+0.75LR)	1.00	1.00	1.00
15	SLS: (D+0.75WZ+0.75LR)	1.00	1.00	1.00
16	SLS: (0.6D+WX)	1.00	1.00	1.00
17	SLS: (0.6D+WZ)	1.00	1.00	1.00
18	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
19	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
20	SLS: (D+WX)	1.00	1.00	1.00
21	SLS: (D+WZ)	1.00	1.00	1.00
35	SLS: (DL+W-Z)	1.00	1.00	1.00
36	SLS: 0.6DL+WZ-	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
22	ULS: (1.4 DL)	1.00	1.00	1.00
23	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
24	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
25	ULS: (1.2D+0.80WX+1.6LR)	1.00	1.00	1.00
26	ULS: (1.2D+1.6WZ+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+1.6WX+0.5LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+EX)	1.00	1.00	1.00
30	ULS: (1.2D+EZ)	1.00	1.00	1.00
31	ULS: (0.90D+1.6WX)	1.00	1.00	1.00
32	ULS: (0.90D+1.6WZ)	1.00	1.00	1.00
33	ULS: (0.90D+EZ)	1.00	1.00	1.00
34	ULS: (0.90D+EX)	1.00	1.00	1.00
37	ULS: (0.9DL+1.6WZ-VE)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
8	226.746	0.000	0.145	0.598	0.000
9	224.142	-0.005	0.095	0.322	0.004
10	236.735	0.000	0.209	0.631	0.000
11	234.238	0.000	0.193	0.623	0.000
12	226.718	0.000	3.638	9.921	0.000
13	226.746	6.411	0.145	0.598	-8.911
14	234.249	11.078	-11.066	-15.730	-13.531
15	234.131	0.000	12.891	21.166	0.000
16	136.062	14.771	-14.925	-21.444	-18.042
17	135.906	0.000	17.018	27.751	0.000
18	136.048	6.411	0.087	0.359	-8.911
19	136.020	0.000	3.580	9.681	0.000
20	226.761	14.771	-14.867	-21.204	-18.042
21	226.604	0.000	17.076	27.990	0.000
35	226.746	0.000	0.145	0.598	0.000
36	136.048	0.000	0.087	0.359	0.000

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
22	317.444	0.000	0.203	0.838	0.000
23	318.793	-0.007	0.165	0.466	0.006
24	318.793	-0.007	0.165	0.466	0.006
25	288.089	11.817	-11.734	-16.673	-14.433
26	276.863	0.000	27.296	44.561	0.000
27	277.113	23.634	-23.814	-34.150	-28.866
28	276.863	0.000	27.296	44.561	0.000
29	272.095	9.158	0.174	0.718	-12.730
30	272.056	0.000	5.165	14.036	0.000
31	204.095	23.634	-23.890	-34.346	-28.866
32	203.844	0.000	27.221	44.365	0.000
33	204.032	0.000	5.121	13.856	0.000
34	204.071	9.158	0.131	0.539	-12.730
37	204.071	0.000	0.131	0.539	0.000

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 2.250 m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 1.301 m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.500 m Governing Load Case : # 8

Width (W_2) = 1.500 m Governing Load Case : # 8

Depth (D_2) = 0.400 m Governing Load Case : # 27

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 2.250 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 21.599 kN

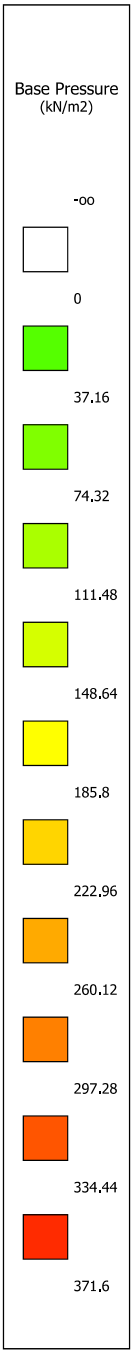
Gross Soil Bearing Capacity = 371.60 kN/m2

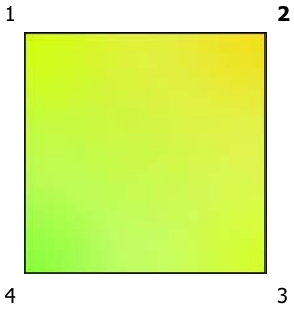
Soil Weight On Top Of Footing = 30.599 kN

Pressures at 4 Corners

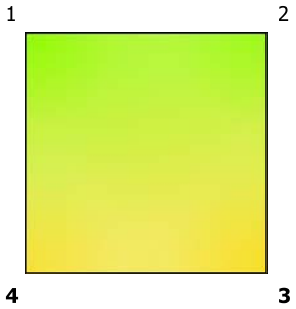
Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner

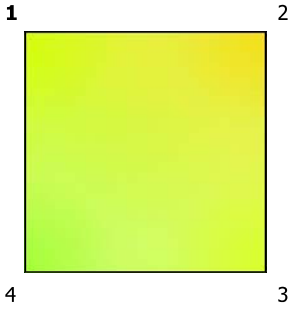




Corner 2

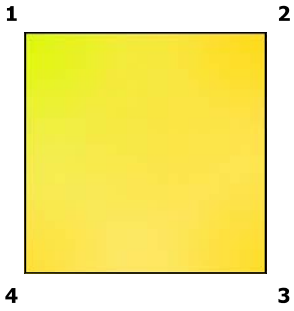


Corner 3, 4



Corner 1

Governing Base Pressure Contour



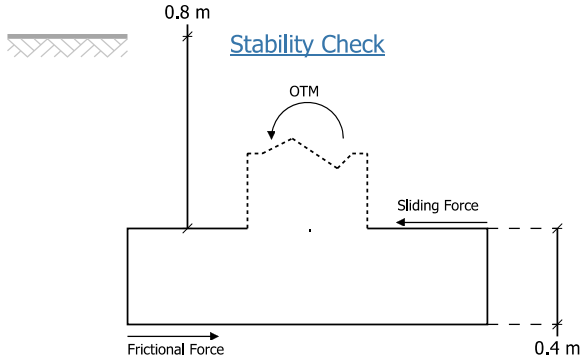
Corner 1, 2, 3, 4

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
14	**131.2098**	195.0772	123.4097	59.5423	0.000
20	129.6731	**214.8296**	118.2902	33.1336	0.000
21	62.0074	62.0074	**185.8167**	185.8167	0.000
21	62.0074	62.0074	185.8167	**185.8167**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
14	**131.2098**	195.0772	123.4097	59.5423
20	129.6731	**214.8296**	118.2902	33.1336
21	62.0074	62.0074	**185.8167**	185.8167
21	62.0074	62.0074	185.8167	**185.8167**



Stability Check

-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
8	N/A	1017.343	1438.741	1.500	320.162	N/A	1.500
9	29380.165	1539.870	2174.720	1.500	579.010	32991.451	1.500
10	N/A	732.610	1036.067	1.500	304.852	N/A	1.500
11	N/A	786.182	1111.830	1.500	308.444	N/A	1.500
12	N/A	40.628	57.457	1.500	18.477	N/A	1.500
13	23.061	1017.343	32.605	1.500	320.162	18.231	1.500
14	13.704	13.719	13.711	1.500	10.709	11.960	1.500
15	N/A	11.772	16.648	1.500	8.197	N/A	1.500
16	6.755	6.685	6.720	1.500	5.187	5.895	1.500
17	N/A	5.858	8.285	1.500	4.112	N/A	1.500
18	15.563	1144.259	22.007	1.500	360.939	12.303	1.500
19	N/A	27.862	39.403	1.500	12.793	N/A	1.500
20	10.009	9.944	9.977	1.500	7.743	8.736	1.500
21	N/A	8.653	12.237	1.500	6.034	N/A	1.500
35	N/A	1017.343	1438.741	1.500	320.162	N/A	1.500
36	N/A	1144.259	1618.226	1.500	360.939	N/A	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 16
Governing Disturbing Force : 14.771kN
Governing Restoring Force : 99.778kN
Minimum Sliding Ratio for the Critical Load Case : 6.755
Critical Load Case for Overturning about X-Direction : 17
Governing Overturning Moment : 34.558kNm
Governing Resisting Moment : 142.088kNm
Minimum Overturning Ratio for the Critical Load Case : 4.112

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 17
Governing Disturbing Force : 17.018kN
Governing Restoring Force : 99.695kN
Minimum Sliding Ratio for the Critical Load Case : 5.858
Critical Load Case for Overturning about Z-Direction : 16
Governing Overturning Moment : -23.950kNm
Governing Resisting Moment : 141.193kNm
Minimum Overturning Ratio for the Critical Load Case : 5.895

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

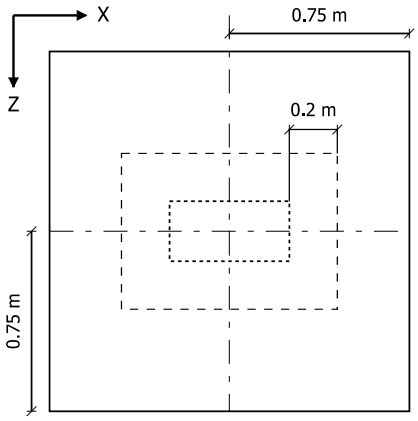
Critical Load Case for Sliding along Resultant Direction : 16
Governing Disturbing Force : 20.999kN
Governing Restoring Force : 141.107kN
Minimum Sliding Ratio for the Critical Load Case : 6.720

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m
Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m
For rectangular column, β_c = B_{col} / D_{col} = 2.000

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 288.328kN, Load Case # 23

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 2.876m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 1944.021kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3297.265kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 1944.021kN

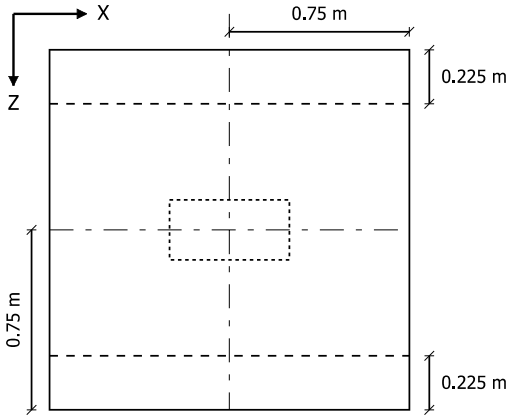
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1458.016kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 506.960kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{zd}$ = 0.281m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

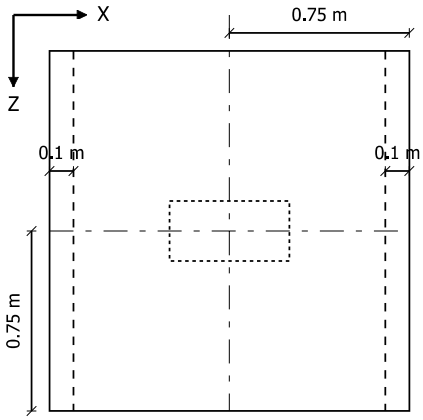
From above calculations, 0.75 X V_c = 380.220 kN

Critical load case for V_{ux} is # 26 $V_{ux} = V_{ux}|_{x=D_x}$ = 95.429 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 506.960 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.156 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

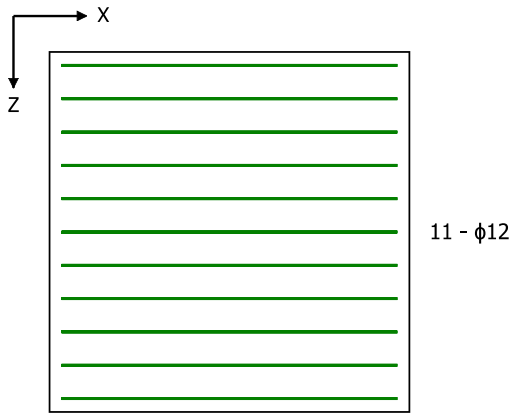
From above calculations, 0.75 X V_c = 380.220 kN

Critical load case for V_{uz} is # 27 $V_{uz} = V_{uz}|_{z=D_z}$ = 48.532 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 27

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488
From ACI Cl. 7.12.2, ρ_{min} =		0.00169
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	1.000 m
Ultimate moment,	$M_u\big\vert_{z=D_z}$ =	37.395 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	41.550 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00056
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00048
Since	ρ= ρ_{min}	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1014.827 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

According to ACI 318 Clause No- 10.6.4 1st

Max spacing for Cracking Consideration = 250.272mm

Safe for Cracking Aspect.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover}$ = 0.450 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 11

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1244.062 mm2

d_{eff} = D - C_{cover} - 0.5 X (dia. of one bar) = 0.344 m

Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00241

From ACI Cl.7.6.1, minimum req'd clear distance between bars

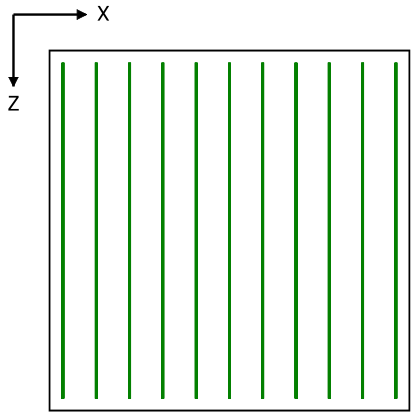
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.226

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



11 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 26

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} = 0.332 m

Factor β_1 from ACI Cl.10.2.7.3 = 0.796

From ACI Cl. 10.3.2, ρ_{bal} = $0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$ 0.03317

From ACI Cl. 10.3.3, ρ_{max} = $0.75 \times \rho_{bal}$ = 0.02488

From ACI Cl.7.12.2, ρ_{min} = 0.00174

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{(0.85 \times F_c')}$ = 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 0.875 m

Ultimate moment, $M_u|_{x=D_x}$ = 63.732 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 70.814 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{(F_y \times W \times d_{eff}^2)}}\right]$ = 0.00103

(Based on gross depth) ρ x d_{eff} / Depth = 0.00085

Since $\rho = \rho_{min}$ ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 1041.114 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

According to ACI 318 Clause No- 10.6.4 3rd

Max spacing for Cracking Consideration = 250.272mm

Safe for Cracking Aspect.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times (L - D_{col}) \cdot - C_{cover}$ = 0.575 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 11

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1244.062 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.332 m

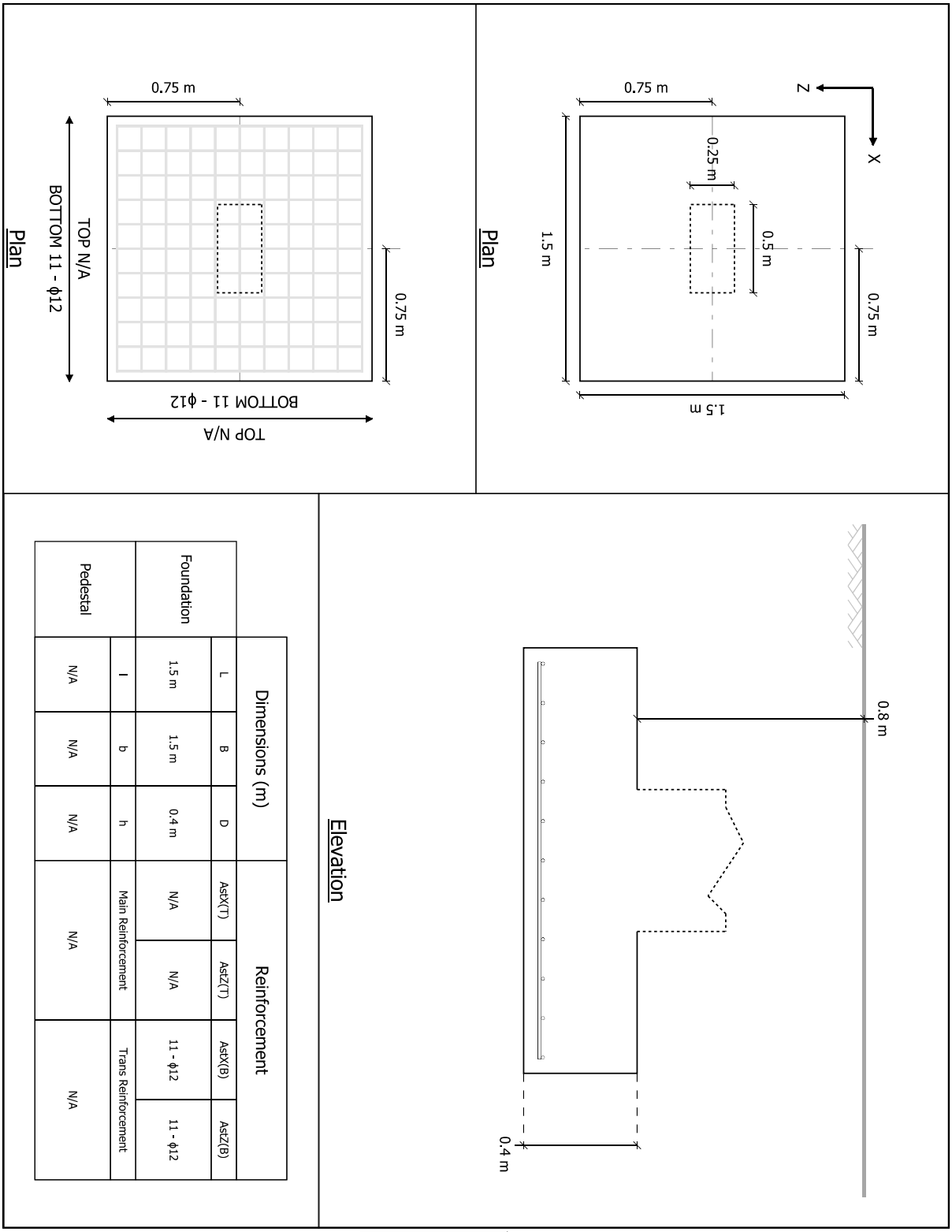
Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00250

From ACI Cl.7.6.1, minimum req'd clear distance between bars

C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.195

Check to see if width is sufficient to accomodate bars



Dimensions (m)			Reinforcement			
L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
1.5 m	1.5 m	0.4 m	N/A	N/A	11 - ϕ12	11 - ϕ12
l	b	h	Main Reinforcement		Trans Reinforcement	
N/A	N/A	N/A	N/A		N/A	

Foundation

Pedestal

Print Calculation Sheet